

Homework 6

For all stability table (Routh array) problems, you can either do them by hand or use the function `routh.m` posted on Canvas. Look at the example in the code for how to call the function correctly by typing “help routh” with the function in your working path.

Dynamic Responses (Chapter 3)

Do problems:

- # 53.
- # 56.
- # 58. Note that the characteristic equation is the equation that has the poles as solutions ($D(s) = 0$). For all parts, consider only positive values of T (but both positive and negative values of A).
 - For part b), use both approximations and compare results.
 - Graphically show the region of stability (shade the region in the T - A plane where the system is stable for each approximation). Explain your methodology.
 - Show the true (exact) step response for different values of T and A (using Simulink or Matlab) without using the approximations. Note that e^{-sT} is a “transport delay”. Pay particular attention to where the approximations result in different stability conclusions.
- # 59.
 - I believe there is a typo in the book problem in some editions that makes this problem hard to solve (the s^0 term should be $5k+1$ not $5k$). You can assume the polynomial $p^3 + (3+k)p^2 + (4k-4)p + 1 = 0$. (characteristic equation after substituting $s = p-1$).

First Analysis of Feedback (Chapter 4)

Do problems:

- # 4. Hint: for the closed loop transfer function T , sensitivity is defined as $\frac{A}{T} \left(\frac{dT}{dA} \right)$, $\frac{a}{T} \left(\frac{dT}{da} \right)$, or $\frac{\beta}{T} \left(\frac{dT}{d\beta} \right)$ for each part, respectively. Read section 4.1.4 of the textbook for more details. You can use the Matlab symbolic toolbox to help with the algebra if necessary.
- # 7c only. Use Simulink. Hint: K_1 will be in the 1000-1500 range to obtain the desired step overshoot, and the steady state error to $r = 5t$ should be around 0.3.
- # 10. For part (b), you can use Matlab function `routh.m` found at: <https://www.mathworks.com/matlabcentral/fileexchange/58-routh-m>
- # 29. Hint: use the Matlab symbolic toolbox to make the algebra trivial.

Root-Locus Design Method (Chapter 5)

Do the following book problems. Note that you can use the Matlab function `rlocus()` for any of the problems below, but make sure you have the correct $L(s)$ when calling `rlocus(L)`. For any asymptote or angle calculations, consider coding in Matlab to make the calculations easier:

1. # 3. Plot using Matlab. Calculate the asymptotes and label them on the plot.
2. # 7. Plot using Matlab. Then, calculate the arrival angles, departure angles, and any asymptotes and label them on your plots. Hint: You can use Matlab to make the repetitive calculations easier.
3. # 8(d) using Matlab. Also specify the value of K for which the maximum damping occurs.
4. # 9
5. # 13 using Matlab
6. # 21 (you can use the `routh.m` function)

Optional bonus:

7. # 18. First test the claim in Matlab for various values of z and p , then prove analytically.